

Rational complexity of binary strings: exact mean, finite distribution and logarithmic variance

Oleksiy Babanskyy

Abstract

For a residue r modulo 2^N , let $R_N(r)$ be the least height $\max(|a|, b)$ of a rational representation $a \equiv rb \pmod{2^N}$ with positive odd denominator b . We prove that the uniform mean of the ordinary, unrounded quantity $\log_2 R_N(r)$ is at most $N/2$ for every $N \geq 1$. This is the mean part of Question 3.1 in the 2026 survey of Lin, Xiao and Chen. The proof compares the product of the two lift heights with a truncated logarithmic excess at the parent modulus. Short lattice vectors with even denominator cover the large-height residues; completion in the actual odd coset bounds their contribution. A finite lattice sum, a two-prime sieve and an explicit primitive-point estimate close the inequality for parent length at least 26. Integer certificates cover the remaining parent lengths $1, \dots, 25$. We also derive the complete finite distribution, hence an exact finite arithmetic-logarithmic formula for the logarithmic variance. A marked-lattice limit identifies the normalized height with a published Boca–Gologan law and yields the limiting logarithmic variance $0.892302571204274625294755\dots$. The raw-height mean and limiting Jensen gap are evaluated. A batched histogram algorithm uses $O(N2^N)$ arithmetic operations and $O(2^N)$ integer storage slots.

1 The problem and the results

Write a binary string as $s = (s_0, \dots, s_{N-1})$, and identify it with $r = \sum_{i=0}^{N-1} s_i 2^i$ modulo 2^N . Define

$$R_N(r) = \min_{\substack{a \equiv rb \pmod{2^N} \\ b > 0, b \text{ odd}}} \max(|a|, b), \quad \mathcal{E}_N = 2^{-N} \sum_{r=0}^{2^N-1} \log_2 R_N(r). \quad (1.1)$$

This is the rational height underlying ordinary finite 2-adic complexity. The logarithm is not rounded to an integer. The measure includes every string, including the zero string; it is not a measure on odd residues.

Theorem 1.1. *For every integer $N \geq 1$,*

$$\mathcal{E}_N \leq \frac{N}{2}.$$

Tian and Qi studied this same mean and proved a weaker upper bound [7, Theorem 3]. Chen and Winterhof proved the lower bound $\mathcal{E}_N \geq N/2 - 1$ and an upper bound with a logarithmic error [8, Theorems 2–3]. Their short primitive counts credit Tian–Qi; their growth recurrence credits Goresky–Klapper [12]. The exact upper in Theorem 1.1 is explicitly asked for in [10, Question 3.1], together with a separate variance problem. We give the exact mean bound, a finite formula for the logarithmic variance, and its evaluated limit. The finite formula is a bounded arithmetic sum involving logarithms, not a short elementary expression in N .

The exact-mean proof uses elementary geometry of congruence lattices and finite integer certificates. It does not use a limiting distribution to infer an inequality at every length. Its computational premises are stated in Section 6 and checked by the accompanying dependency-free program. The uniform argument is a proof for all sufficiently large lengths, not an extrapolation of those certificates.

Sections 7–8 give the complete finite distribution, moment sums and a batched histogram algorithm. For $Z_N = R_N/2^{N/2}$, Sections 9–10 prove a marked-lattice limit and identify its survival as $G_2(t^2)$, where G_2 is the published Boca–Gologan law. Section 11 evaluates its logarithmic moments and transfers them to binary strings:

$$\mathcal{E}_N = N/2 + \frac{7\zeta(3)/(2\pi^2) - 1}{2\ln 2} + o(1), \quad \text{Var}(\log_2 R_N) \longrightarrow 0.892302571204274625294755 \dots$$

Section 12 evaluates the leading raw-mean coefficient and the positive limiting Jensen gap. The law itself is known; the dyadic identification, finite formulas and moment evaluations are stated separately, with their external hypotheses at the point of use.

Source and priority boundary. For Chen–Winterhof’s detailed statements we use the explicitly identified arXiv version 1 [8]. The final publication [9] has been identified, but its full body has not been obtained for this paper. We do not assert a lemma-by-lemma identity of those versions. The July 2026 survey, which includes Chen as an author, explicitly poses the exact mean question after the final publication. No exact antecedent has been identified in the primary sources checked for this paper; this is not a certification of universal priority. Ren–Winterhof’s recent symmetric complexity [11] minimizes also over a reversed string and is a different random variable.

Notation. Logarithms without a displayed base are natural; finite-string moments use base two. The subscript in R_N denotes a length. When a parent modulus is $M = 2^n$, abbreviate $R_M(r) = R_n(r)$; thus R_{2M} denotes the child height R_{n+1} . Put

$$q = \sqrt{M}, \quad s = q/\sqrt{2}, \quad J = \lfloor n/2 \rfloor.$$

The convention $[x]_+ = \max(x, 0)$ will be used. A minimum in (1.1) is primitive in $\mathbb{Z}^2$: its common divisor is odd and invertible modulo 2^N , so division would otherwise lower its height. In particular $R_M(0) = 1$ and $R_M(r) \leq M/2$ for $M \geq 2$.

2 The two lifts and a logarithmic budget

Lemma 2.1. *Let $M = 2^n$, $n \geq 1$. For a residue r modulo M , put*

$$h = R_M(r), \quad H = \max\{R_{2M}(r), R_{2M}(r + M)\}.$$

Then $\min\{R_{2M}(r), R_{2M}(r + M)\} = h$, and

$$hH \leq \max(M, 2h^2). \tag{2.1}$$

Proof. The union of the two sets of child representations is exactly the set of parent representations, since an odd denominator is invertible modulo $2M$. Choose a primitive parent minimum $u = (a, b)$, $b > 0$ odd. It selects a child r_0 with height h . Reflection allows $a \geq 0$.

Suppose first $a > 0$. Integer solutions $v = (c, d)$ of $ad - bc = M$ form $v_0 + ku$ by Bézout’s identity. The restriction d odd selects exactly one class of k modulo 2. Moreover

$$c \equiv (r_0 + M)d \pmod{2M}.$$

Indeed $a \equiv r_0 b \pmod{2M}$ and $b^{-1}M \equiv M \pmod{2M}$, while $Md \equiv M$ for odd d . A negative d can be oriented positive by negating v ; its height and child are unchanged.

Put $\tau = \min(a, b)$, $h = \max(a, b)$, $\sigma = a + b$ and $X = \max\{h, (M + 2ab)/\sigma\}$. Parameterize the real determinant line by $(-M/\sigma, M/\sigma) + t(a, b)$. If $M \leq h(h - \tau)$, then $X = h$. When $a = h$, the full interval $c \in [-h, h]$ gives $|d| = |(M + bc)/a| \leq h$; its length in t is 2. When $b = h$, use

the full interval $d \in [-h, h]$ instead. If $M \geq h(h - \tau)$, then $X = (M + 2ab)/\sigma \geq h$. The points at $t = -2b/\sigma$ and $t = 2a/\sigma$ are

$$\left(-X, \frac{M - 2b^2}{\sigma}\right), \quad \left(\frac{-M + 2a^2}{\sigma}, X\right).$$

Both belong to $[-X, X]^2$: the required conditions are $M \geq b(b - a)$ and $M \geq a(a - b)$, which follow from the case assumption. The closed segment between them has parameter length 2. In either case it intersects the class $t_0 + 2\mathbb{Z}$. Thus the other child has height at most X .

Writing $x = \tau/h$, we obtain

$$hH \leq \max \left\{ h^2, \frac{M + 2h^2x}{1 + x} \right\} \leq \max(M, 2h^2).$$

If $a = 0$, primitivity forces $b = 1$, the children are 0 and M , and their heights are 1 and M , so the same conclusion holds. Ties between parent minima need no special choice. $\square$

Define the exact finite quantities

$$\begin{aligned} C_M(t) &= \#\{r \bmod M : R_M(r) \leq t\}, \\ K_M &= \frac{1}{M} \sum_r [\ln(R_M(r)/s)]_+, \\ \mathcal{D}_M &= \frac{1}{M} \int_s^q C_M(t) \frac{dt}{t}, \\ \mathcal{L}_M &= \frac{1}{M} \sum_r [\ln(R_M(r)/q)]_+. \end{aligned} \tag{2.2}$$

We reserve D without a calligraphic letter for a dyadic content below. Taking logarithms in (2.1), and averaging over both children, gives

$$\mathbb{E}_{r \bmod 2M} \ln R_{2M}(r) \leq \frac{1}{2} \ln M + K_M. \tag{2.3}$$

There is no induction hypothesis on the parent mean.

For every $R > 0$,

$$[\ln(R/s)]_+ = \ln(q/s) - \int_s^q \mathbf{1}_{R \leq t} \frac{dt}{t} + [\ln(R/q)]_+.$$

Consequently

$$K_M = \frac{1}{2} \ln 2 - \mathcal{D}_M + \mathcal{L}_M. \tag{2.4}$$

It suffices to prove $K_M \leq \frac{1}{2} \ln 2$ at every parent length. For large lengths we prove $\mathcal{L}_M < \mathcal{D}_M$.

3 Short even vectors and the actual odd coset

For $r \bmod M$ let

$$\Lambda_r = \{(A, B) \in \mathbb{Z}^2 : A - rB \in M\mathbb{Z}\}.$$

Its basis $(M, 0), (r, 1)$ has determinant M .

Lemma 3.1. *If $R_M(r) > q$, there exist*

$$D = 2^j, \quad 1 \leq j \leq J, \quad b > 0 \text{ odd}, \quad \gcd(a, b) = 1, \quad D \max(|a|, b) \leq q$$

such that

$$a - rb = (M/D)\alpha_r, \quad \alpha_r \text{ odd}. \tag{3.1}$$

For each such triple, its cell $\mathcal{C}(D, a, b)$ of residues satisfying (3.1) has exactly $D/2$ elements. For each residue in that cell, $D(a, b)$ is primitive as an element of Λ_r .

Proof. Minkowski's convex body theorem [13] in the square of radius $q + \varepsilon$ gives a nonzero lattice vector of height at most $q + \varepsilon$. Discreteness permits passage to the closed radius q . Choose a shortest nonzero vector. Its denominator is nonzero, since a nonzero horizontal vector has height at least $M > q$. Orient it positive. The hypothesis $R_M(r) > q$ forces its denominator even, and its numerator is then even as well. A common odd factor could be divided out while preserving lattice membership, contradicting minimality. Thus its content is $D = 2^j \geq 2$; write it as $D(a, b)$ with $\gcd(a, b) = 1$. The congruence modulo $Q = M/D$, which is even, forces b odd. If $\alpha_r = (a - rb)/Q$ were even, division by 2 would again yield a shorter vector of Λ_r . This proves the asserted contact.

There is one solution of $a - rb \equiv 0 \pmod{Q}$. Its D lifts modulo M change α_r by $-bt$, $t \pmod{D}$. Exactly $D/2$ have odd contact. The coordinates of $D(a, b)$ in the displayed lattice basis are (α_r, Db) . Their gcd is 1: α_r is odd, and $\gcd(\alpha_r, b) = 1$ follows from $\gcd(a, b) = 1$ and $\gcd(Q, b) = 1$. This is lattice primitivity, despite the even integer content. $\square$

Lemma 3.2. *On a cell $\mathcal{C}(D, a, b)$ put $A = D|a|$, $B = Db$. Then*

$$R_M(r) \leq X := \frac{M + AB}{A + B}, \quad X \geq q. \quad (3.2)$$

On the axis $a = 0, b = 1$, the height is exactly $R_M(r) = M/D$.

Proof. For $a > 0$, lattice primitivity allows completion of $v = (A, B)$ to a basis with determinant M . Such completions $w = (c, d)$ satisfy

$$ad - bc = Q, \quad ad \equiv Q \pmod{b}, \quad \alpha_r d \equiv 1 \pmod{D}.$$

The last two congruences give one class of odd d modulo $B = Db$ by the Chinese remainder theorem. Invert a only modulo b ; no inversion modulo an even denominator is needed. For $b = 1$ the first congruence is empty. All completions differ by a multiple of v .

Put $x = M/(A + B)$. Let $d_- \leq x < d_+ = d_- + B$ be the adjacent denominators in this class, and write $\eta = x - d_- \in [0, B)$. On the determinant line, $c(d) = (Ad - M)/B$. The two heights are

$$\max(|c(d_-)|, |d_-|) = x + (A/B)\eta, \quad \max(|c(d_+)|, |d_+|) = x + B - \eta. \quad (3.3)$$

For clarity, the sign checks use $A, B \leq q$. If $d_- < 0$ and $A < B$, then $(B - A)|d_-| \leq (B - A)(B - x) \leq B^2 \leq M$, so $-c(d_-) \geq |d_-|$; the case $A \geq B$ is immediate. At $d_+ > x$, $-c(d_+) \leq d_+$. If $A > B$, then $c(d_+) \leq d_+$ follows from $(A - B)(x + B) \leq M$, equivalent to $A^2 - B^2 \leq 2M$. The other sign cases follow directly from $d_- \leq x < d_+$. The smaller expression in (3.3) is at most $x + AB/(A + B) = X$, since the two affine expressions meet at $\eta = B^2/(A + B)$. Negative odd denominators are oriented positive by negating both coordinates. Reflection handles $a < 0$. Finally $X/q \geq 1$ is equivalent to $(1 - A/q)(1 - B/q) \geq 0$.

If $a = 0$, primitivity gives $b = 1$ and $r = -Q\alpha_r$. Every marked numerator is a nonzero odd multiple of Q , so the height is at least Q . Take $c = -Q$ and a positive odd $d \leq D - 1$ representing $\alpha_r^{-1} \pmod{D}$. This has height Q , because $D \leq q \leq Q$. $\square$

Remark 3.3. Only coverage and the cell masses are needed for our upper bound. We do not require disjointness of the cells or replace their actual odd phases by continuous phase averages. The union bound below remains valid even without a uniqueness statement.

Let $V = \{r \neq 0 : v_2(r) \geq \lceil n/2 \rceil\}$. The axis cells are precisely these residues: for $D = 2^j$ their valuation is $k = n - j$ and their height is 2^k . This family is already present in Tian–Qi [7, Lemma 4]. Write $\mathcal{L}_V$ for its contribution to $\mathcal{L}_M$ and $\mathcal{L}_{\text{other}} = \mathcal{L}_M - \mathcal{L}_V$. For $0 < x, y \leq 1$ define

$$f(x, y) = \ln \frac{1 + xy}{x + y}, \quad P_f(T) = \sum_{\substack{1 \leq a, b \leq T \\ b \text{ odd}, \gcd(a, b) = 1}} f(a/T, b/T).$$

The kernel is nonnegative. Every non-axis residue with height above q is covered by Lemma 3.1, and Lemma 3.2 bounds its energy by $f(D|a|/q, Db/q)$. Each cell has $D/2$ residues and both signs of a give the weight D . Therefore, using an upper bound by a union of cells,

$$\mathcal{L}_{\text{other}} \leq U_M := \frac{1}{M} \sum_{j=1}^J 2^j P_f(q/2^j). \quad (3.4)$$

Including cells whose height is at most q only enlarges this upper bound; all summands are nonnegative.

4 An explicit estimate of the lattice sum

Let

$$I = \int_{[0,1]^2} f(x, y) dx dy, \quad F(T) = \sum_{\substack{1 \leq a, b \leq T \\ b \text{ odd}}} f(a/T, b/T).$$

The integral is finite: $0 \leq f(x, y) \leq -\ln \max(x, y)$, so $I \leq 1/2$.

Lemma 4.1. *For every real $T > 0$,*

$$\frac{I}{2} T^2 - T \leq F(T) \leq \frac{I}{2} T^2, \quad I = \frac{\zeta(2) - 1}{2} = \frac{\pi^2}{12} - \frac{1}{2}. \quad (4.1)$$

Proof. For fixed $0 < x \leq 1$, extend $f(x, y)$ to $y \geq 0$. Its derivatives satisfy

$$\partial_y f = \frac{x^2 - 1}{(1 + xy)(x + y)} \leq 0, \quad \partial_y^2 f = \frac{1}{(x + y)^2} - \frac{x^2}{(1 + xy)^2} \geq 0.$$

The last assertion follows from $1 + xy \geq x(x + y)$, also beyond $y = 1$. Odd $b \leq T$ are the midpoints of intervals $[(b - 1)/T, (b + 1)/T]$. Convexity gives

$$\frac{2}{T} \sum_{b \leq T, b \text{ odd}} f(x, b/T) \leq \int_0^{Y_T} f(x, y) dy, \quad Y_T = \frac{2\lceil T \rceil / 2}{T}.$$

If $Y_T \leq 1$, the omitted portion is nonnegative. If $Y_T > 1$, the added portion is nonpositive, because $f(x, y) \leq 0$ for $y > 1$. Thus this integral is at most $\Phi(x) = \int_0^1 f(x, y) dy$. The function Φ is decreasing and nonnegative, so its right Riemann sum at a/T , $1 \leq a \leq \lceil T \rceil$, is at most TI . This proves the upper bound for real T , including the last odd cell.

For $T \geq 1$, the clipped rectangles $[a/T, (a + 1)/T] \times [b/T, (b + 2)/T]$ with the same indices cover $[1/T, 1]^2$ with disjoint interiors. Monotonicity bounds the integral there by $2F(T)/T^2$. The omitted vertical and horizontal strips have total integral at most $2/T$, using $f \leq -\ln y$ and $f \leq -\ln x$, respectively. This proves the lower bound. For $0 < T < 1$, $F(T) = 0$ and the lower bound is nonpositive.

For $z \geq 1$, the area of the region $(1 + xy)/(x + y) \geq z$ is

$$\mathcal{A}(z) = \int_0^{1/z} \frac{1 - zx}{z - x} dx = 1 - (z^2 - 1) \ln \frac{z^2}{z^2 - 1} = \sum_{k \geq 1} \frac{z^{-2k}}{k(k + 1)}.$$

The endpoint $z = 1$ is interpreted by a limit. Layer cake and Tonelli's theorem give

$$I = \int_1^\infty \mathcal{A}(z) \frac{dz}{z} = \frac{1}{2} \sum_{k \geq 1} \frac{1}{k^2(k + 1)} = \frac{\zeta(2) - 1}{2}.$$

The last expression uses $1/(k^2(k + 1)) = 1/k^2 - 1/k + 1/(k + 1)$ and Euler's classical evaluation $\zeta(2) = \pi^2/6$. $\square$

A primitive pair cannot have either 3 or 5 as a common divisor. Since $f \geq 0$, inclusion-exclusion in the larger set which excludes only those two common divisors gives

$$P_f(T) \leq F(T) - F(T/3) - F(T/5) + F(T/15).$$

Dividing by an odd common divisor preserves the odd denominator. Using the upper bound in (4.1) in the positive terms and the lower bound in the negative terms yields

$$P_f(T) \leq \frac{32I}{75}T^2 + \frac{8}{15}T.$$

This includes scaled radii smaller than 1. Substitution in (3.4) gives the uniform finite estimate

$$\mathcal{L}_{\text{other}} \leq \frac{32I}{75}(1 - 2^{-J}) + \frac{8J}{15q}. \quad (4.2)$$

5 Central credit and the infinite range

For integer $T \geq 1$, define the signed primitive count, with the axis once,

$$A(T) = \#\{(a, b) : |a| \leq T, 1 \leq b \leq T, b \text{ odd}, \gcd(a, b) = 1\}.$$

Lemma 5.1. *For every integer $T \geq 1$,*

$$A(T) \geq \frac{8}{\pi^2}T^2 - \frac{3}{4}T \ln T - \frac{11}{4}T - \frac{1}{4}. \quad (5.1)$$

For $n \geq 2$, $T_0 = \lfloor q/\sqrt{2} \rfloor$ satisfies $C_M(T_0) = A(T_0)$, and

$$\mathcal{D}_M \geq \frac{2 \ln 2}{\pi^2} - \frac{4\sqrt{2} \ln 2 / \pi^2 + (\ln 2 / (2\sqrt{2}))((3/4) \ln q + 11/4)}{q} - \frac{\ln 2}{8q^2}. \quad (5.2)$$

Proof. Möbius inversion over odd common divisors gives the exact formula

$$A(T) = \sum_{\substack{g \leq T \\ g \text{ odd}}} \mu(g)(2\lfloor T/g \rfloor + 1) \left\lceil \frac{\lfloor T/g \rfloor}{2} \right\rceil. \quad (5.3)$$

For $l = \lfloor x \rfloor$ the raw count $B(l) = (2l+1)\lfloor l/2 \rfloor$ equals $l^2 + l/2$ when l is even and $l^2 + 3l/2 + 1/2$ when it is odd. Hence $|B(l) - x^2| \leq 3x/2 + 1/2$. Also

$$\sum_{\substack{g \leq T \\ g \text{ odd}}} \frac{1}{g} \leq 1 + \frac{1}{2} \ln T, \quad \#\{g \leq T : g \text{ odd}\} \leq (T+1)/2.$$

For the first inequality, $1/(2k+1) \leq \frac{1}{2} \ln((2k+1)/(2k-1))$ for $k \geq 1$ telescopes. The raw error in (5.3) is at most $(3/4)T \ln T + (7/4)T + 1/4$. The absolutely convergent Euler product gives $\sum_{g \text{ odd}} \mu(g)/g^2 = 8/\pi^2$; truncating it at T costs at most $T^2 \sum_{g > T} g^{-2} \leq T$. This proves (5.1).

Two distinct primitive representations in the T_0 box of one residue have a nonzero determinant divisible by M , unless they coincide. The determinant has absolute value at most $2T_0B_0$, where B_0 is the largest odd integer at most T_0 . If $2T_0^2 < M$, this is less than M . If $2T_0^2 = M$ and $n \geq 2$, T_0 is a power of 2 and is even, so $B_0 = T_0 - 1$ and the strict inequality still holds. A zero determinant gives the same primitive fraction with positive denominator. Thus the residue map is injective; every covered residue has a primitive representation in the box, so $C_M(T_0) = A(T_0)$. The excluded case $n = 1$ has the collision $(1, 1), (-1, 1)$.

By monotonicity $\mathcal{D}_M \geq (\ln 2)/(2M)A(T_0)$. Use $T_0^2 \geq q^2/2 - \sqrt{2}q$, $T_0 \ln T_0 \leq (q/\sqrt{2}) \ln q$ and $T_0 \leq q/\sqrt{2}$ in (5.1). This yields (5.2). $\square$

The axis mass at valuation k is 2^{-k-1} . Consequently

$$\frac{\mathcal{L}_V}{\ln 2} = \sum_{k=\lceil n/2 \rceil}^{n-1} 2^{-k-1}(k - n/2) = \begin{cases} 2^{-\ell} - (\ell + 1)2^{-2\ell}, & n = 2\ell, \ell \geq 1, \\ 3 \cdot 2^{-\ell-2} - (2\ell + 3)2^{-2\ell-2}, & n = 2\ell + 1, \ell \geq 0. \end{cases} \quad (5.4)$$

In particular $\mathcal{L}_V \leq \sqrt{2} \ln 2 / q$ for all $n \geq 1$. These are finite geometric sums, including the zero energies at $n = 1, 2$.

Proposition 5.2. *For every parent length $n \geq 26$, $\mathcal{L}_M < \mathcal{D}_M$, hence $K_M < \frac{1}{2} \ln 2$.*

Proof. Put

$$c = \frac{32I}{75} = \frac{16}{75} \left(\frac{\pi^2}{6} - 1 \right), \quad \delta = \frac{2 \ln 2}{\pi^2} - c.$$

Equations (4.2), (5.2), (5.4), discarding the favorable term $-c2^{-J}$, give $\mathcal{D}_M - \mathcal{L}_M \geq \delta - E(q, J)$, where

$$E(q, J) = \frac{8J/15 + 4\sqrt{2} \ln 2 / \pi^2 + (\ln 2 / (2\sqrt{2}))((3/4) \ln q + 11/4) + \sqrt{2} \ln 2}{q} + \frac{\ln 2}{8q^2}.$$

Use $3 < \pi < 22/7$, $1 < \sqrt{2} < 3/2$, and $\ln 2 > 2/3 + 2/81 = 56/81$ from the positive atanh series. Then

$$\delta > \frac{1372}{9801} - \frac{304}{2205} = \frac{45756}{21611205} > \frac{1}{500}.$$

Since $q \geq 2^J$ and $\ln q < J + 1$, the same elementary bounds give

$$E(q, J) \leq \left(\frac{109}{120} J + \frac{47}{12} \right) 2^{-J} + \frac{1}{8} 2^{-2J}.$$

At $J = 13$ the right side is $629/327680 + 1/536870912 < 1/520 + 1/13000 = 1/500$. Both terms decrease for $J \geq 13$; for the first, $aJ + b > a$ with $a = 109/120$, $b = 47/12$. Thus $n \geq 26$ implies $E < 1/500 < \delta$. $\square$

6 Finite certificates and completion of the proof

The remaining premise is $K_{2^n} \leq \frac{1}{2} \ln 2$ for $1 \leq n \leq 25$. There are two different finite interfaces, described here so the computation has an exact mathematical meaning. No decision uses floating logarithms.

For $n = 1, \dots, 12$, certificates give the height h_r and a primitive witness (a_r, b_r) for every residue. The checker verifies $b_r > 0$ odd, $a_r \equiv rb_r$, and $\max(|a_r|, b_r) = h_r$. To prove minimality, for every positive odd $b < h_r$ it computes the least absolute representative of $rb \bmod M$ and checks that it is at least h_r . This excludes every improving pair. Let $c_n = \#\{r : 2h_r^2 > M\}$ and $P_n = \prod_{2h_r^2 > M} h_r^2$. Then the required comparison is exactly

$$P_n \leq 2^{M+(n-1)c_n}. \quad (6.1)$$

This follows by exponentiating $2MK_M \leq M \ln 2$. The full domain has $\sum_{n=1}^{12} 2^n = 8190$ minima. For diagnostic support, the checker also verifies $\prod_r h_r^2 \leq M^M$, but (6.1) is the premise used with the child lemma.

For $n = 13, \dots, 25$, set $T_0 = \lfloor \sqrt{M/2} \rfloor$ and compute $A = A(T_0)$ exactly by (5.3). Put

$$V_2 = \sum_{\substack{0 \leq k < n \\ 2k > n}} 2^{n-k-1}(2k - n) = \frac{2M\mathcal{L}_V}{\ln 2}, \quad x_{D,a,b} = \frac{(M + D^2ab)^2}{MD^2(a+b)^2} \geq 1.$$

Thus $2MU_M = \sum_{D,a,b} D \ln x_{D,a,b}$, with $D = 2^j$, $1 \leq j \leq J$, $1 \leq a, b \leq q/D$, b odd, $\gcd(a, b) = 1$. Using $S = 2^{40}$, let $\ell^+(x) \geq S \ln x$ and $\ell_2^- \leq S \ln 2$ be the integer bounds of Appendix A. The sufficient certificate inequality is

$$\sum_{D,a,b} D \ell^+(x_{D,a,b}) \leq (A - V_2) \ell_2^-, \quad A > V_2. \quad (6.2)$$

Indeed (6.2) implies $U_M + \mathcal{L}_V \leq A \ln 2 / (2M) \leq \mathcal{D}_M$. The integer margin, right side minus left side, is listed below. It is a scaled certificate margin, not the exact real difference $\mathcal{D}_M - \mathcal{L}_M$.

Parent n	Integer margin	Parent n	Integer margin
13	200913119387100	20	22717214856196764
14	343712368064520	21	45370620212212808
15	778422546045060	22	90744127844519936
16	1668697844541872	23	181329491713437076
17	2824453720758008	24	359749179298450700
18	5916038348384600	25	720718942077388912
19	11572174724235024		

The fixed panel contains 9 068 656 positive primitive pairs, out of 11 182 008 raw pairs. The input and replay map in the companion separate these essential certificates from later diagnostic controls.

Proof of Theorem 1.1. The small certificates prove the sufficient bound on K_M for parents $n = 1, \dots, 12$; the directed-log certificates prove it for $n = 13, \dots, 25$. Proposition 5.2 proves it for $n \geq 26$. Equation (2.3) therefore gives $\mathbb{E} \ln R_N \leq (N/2) \ln 2$ for every $N \geq 2$, using parent length $n = N - 1$. For $N = 1$, both residues have height 1. Divide by $\ln 2$. $\square$

7 The exact finite distribution

In the remaining sections $M = 2^N$, $B = M/2$, and $R = R_N$. Write $C_M(T) = \#\{r : R(r) \leq T\}$ and $S_M(T) = M - C_M(T)$. Thresholds in finite formulas are integers. Set $C_M(0) = 0$ and $C_M(T) = M$ for $T \geq B$; a real threshold is handled by its floor. The conventions include $N = 1$, when both heights are one.

Theorem 7.1. *For $1 \leq T \leq \lfloor \sqrt{M} \rfloor$, put*

$$P(T) = \sum_{\substack{g \leq T \\ g \text{ odd}}} \mu(g) \lfloor T/g \rfloor \left\lceil \frac{\lfloor T/g \rfloor}{2} \right\rceil, \quad (7.1)$$

$$J_M(T) = \sum_{\substack{b, d \leq T \text{ odd} \\ (b, d) = 1}} \left[\left\lfloor \frac{T - a_0}{b} \right\rfloor - \left\lfloor \frac{M - bT - da_0}{bd} \right\rfloor + 1 \right]_+, \quad (7.2)$$

where $0 \leq a_0 < b$, $a_0 d \equiv M \pmod{b}$, and $a_0 = 0$ for $b = 1$. Then

$$C_M(T) = 1 + 2P(T) - J_M(T). \quad (7.3)$$

For $T^2 \geq M$ and $T < B$, take all even q with $2 \leq q < M/T$ and all $1 \leq a < q$ with $(a, q) = 1$. Write $v = a^{-1} \pmod{q}$ in $\{1, \dots, q-1\}$, and set

$$b_L = v + q \left\lfloor \frac{T - v}{q} \right\rfloor, \quad b_R = q - v + q \left\lfloor \frac{T - (q - v)}{q} \right\rfloor, \quad (7.4)$$

$$\alpha_{a,q} = \frac{aMb_L - M + qT}{qb_L}, \quad \beta_{a,q} = \frac{aMb_R + M - qT}{qb_R}. \quad (7.5)$$

Then

$$S_M(T) = \sum_{\substack{2 \leq q < M/T \\ q \text{ even}}} \sum_{\substack{1 \leq a < q \\ (a,q)=1}} [\beta_{a,q}] - [\alpha_{a,q}] - 1]_+. \quad (7.6)$$

The two ranges cover all integer thresholds and agree at a common square-root threshold. In particular,

$$\Pr(R_N = t) = \frac{C_M(t) - C_M(t-1)}{M}, \quad 1 \leq t \leq B. \quad (7.7)$$

Proof. The number of primitive pairs $1 \leq a, b \leq T$, b odd, is $P(T)$ by odd-divisor Möbius inversion. In each strict numerator-sign half, the map $(a, b) \mapsto ab^{-1} \bmod M$ is injective: a nonzero determinant has magnitude at most $TB_T < M$, where B_T is the largest odd integer at most T . If $T^2 = M$, T is even and $B_T = T - 1$. The numerator-zero primitive pair $(0, 1)$ cannot collide with a nonzero numerator. A positive/negative collision is exactly

$$ad + eb = M, \quad 1 \leq a, e \leq T, \quad b, d \leq T \text{ odd}.$$

The determinant is positive and less than $2M$. Every solution is automatically primitive and $(b, d) = 1$, since a common odd divisor would divide M . Writing $a = a_0 + bz$ gives $e = (M - da_0)/b - dz$. The conditions $1 \leq a, e \leq T$ are precisely

$$\frac{M - bT - da_0}{bd} \leq z \leq \frac{T - a_0}{b} :$$

the lower bound forces $a > 0$ and the upper bound forces $e > 0$, because $M > TB_T$. Each sign half has at most one witness per residue, so inclusion–exclusion subtracts each collision once. This proves (7.3).

For the high range, let $\mathcal{F}_T^{\text{odd}}$ be the reduced fractions in $[0, 1]$ with odd denominator at most T , including $0/1, 1/1$. The exact interval model is

$$R_N(r) \leq T \iff \frac{r}{M} \in \bigcup_{p/b \in \mathcal{F}_T^{\text{odd}}} \left[\frac{p}{b} - \frac{T}{Mb}, \frac{p}{b} + \frac{T}{Mb} \right]. \quad (7.8)$$

Indeed $a = rb - Mp$ gives $\gcd(a, b) = \gcd(p, b)$ because $(b, M) = 1$; division by odd content preserves coverage. The endpoint fractions provide the wrap intervals $[0, T]$ and $[M - T, M - 1]$ on the residue grid. No interior fraction wraps, since $T < M/2$.

Here are the required Farey facts with their proof. For reduced a/q with $q \leq T$, take the largest $b \leq T$ satisfying $ab \equiv 1 \pmod{q}$, and $p = (ab - 1)/q$. Then $b > T - q$ and $ab - pq = 1$. A fraction strictly between a determinant-one pair has positive integer coordinates in that pair, hence denominator at least $b + q > T$. Thus p/b is the immediate Farey predecessor; the successor uses $ad \equiv -1 \pmod{q}$. Endpoints follow directly. Consecutive denominators therefore sum to more than T and have determinant one; they cannot both be even. Deleting even denominators leaves pairs that were full Farey neighbors or had exactly one deleted even fraction a/q between them.

Put $\lambda = T/M$, so $\lambda T \geq 1$. Retained full-Farey neighbors with denominators b, d have distance $1/(bd)$, less than $\lambda(1/b + 1/d)$ because $b + d > T$. Their closed intervals overlap. Across a deleted even a/q , the immediate odd neighbor denominators are (7.4) and exceed $T - q$. If $\lambda q \geq 1$ both intervals reach the center, leaving no hole. Hence only $qT < M$ contributes; such q satisfy $q < M/T \leq T$ and occur in the sequence.

When $qT < M$, every retained p/b has determinant distance $\delta = |ab - pq| \geq 1$. Its nearest endpoint on its side is at distance $(\delta - \lambda q)/(qb)$ from the center. Among determinant-one candidates, the largest denominator is best. Every $\delta \geq 2$ candidate, anywhere in the sequence, is worse, because

$$\frac{\delta - \lambda q}{b} \geq \frac{2 - \lambda q}{T} > \frac{1}{T}, \quad \frac{1 - \lambda q}{b_L}, \frac{1 - \lambda q}{b_R} < \frac{1}{T}.$$

The latter follows from $T(1 - \lambda q) \leq T - q < b_L, b_R$. This global comparison excludes hidden coverage by broad intervals from more distant centers. The uncovered component is exactly

$$\left(\frac{a}{q} - \frac{1 - \lambda q}{qb_L}, \frac{a}{q} + \frac{1 - \lambda q}{qb_R} \right).$$

It lies strictly inside its retained Farey gap. These components are disjoint and exhaust the uncovered set; endpoint gaps are already covered. Scaling by M gives (7.5). An open interval (α, β) contains exactly $[\beta] - [\alpha] - 1$ integers. Endpoints are covered, and $qT = M$ gives no hole. This proves (7.6). Both formulas count the original event, so they agree where they overlap. Equation (7.7) follows by subtraction. $\square$

Theorem 7.2. *For $j = 1, 2$, the exact finite logarithmic moments and variance are*

$$m_{j,N} = \frac{1}{M} \sum_{T=1}^{B-1} S_M(T) [(\log_2(T+1))^j - (\log_2 T)^j], \quad (7.9)$$

$$\text{Var}(\log_2 R_N) = m_{2,N} - m_{1,N}^2. \quad (7.10)$$

Every coefficient is given by Theorem 7.1 using finite integer arithmetic, gcds, inverses, floors and ceilings.

Proof. Telescope $(\log_2 h)^j$ from height one for each integer $h \in [1, B]$. An increment at T occurs exactly for residues with $R_N > T$. Averaging proves (7.9); the variance identity follows. For $N = 1$ the empty sums give variance zero. $\square$

These are finite arithmetic-logarithmic sums, not a statement that the variance is rational or has a short elementary formula in N . For $M = 4$, the masses 3, 1 at heights 1, 2 give variance $3/16$. For $M = 16$, masses 3, 2, 6, 2, 2, 1 at heights 1, 2, 3, 4, 5, 8 show gaps in the support. At $T = 4$ the high hole is $(20/3, 28/3)$, containing 7, 8, 9; $qT = M$ produces no hole. The extreme heights at $r = M/2 \pm 1$, namely $M/4 + 1$ for $N > 2$, are already recorded in Tian–Qi [7, Lemma 8, p. 74]; the unique top maximum is also recorded there [7, Lemma 10, p. 77].

8 A batched histogram algorithm

Proposition 8.1. *The histogram of R_N and the coefficients of (7.9) can be assembled in $O(M \log M) = O(N2^N)$ elementary arithmetic operations and $O(M)$ integer storage slots. Logarithm evaluation to a prescribed precision is separate.*

Proof. Put $L = \lfloor \sqrt{M} \rfloor$, $K = \lceil \sqrt{M} \rceil$. The new positive primitive boundary $p_t = P(t) - P(t-1)$ is

$$p_1 = 1, \quad p_t = \phi(t) \ (t \geq 2 \text{ even}), \quad p_t = 3\phi(t)/2 \ (t \geq 3 \text{ odd}).$$

For even t only the numerator boundary enters, and all coprime residues modulo t are odd. For odd t the denominator boundary gives $\phi(t)$; the remaining numerator boundary gives $\phi(t)/2$ by pairing b with $t - b$. The corner is nonprimitive for $t > 1$. A totient sieve supplies these values.

Pass once through the ordered coprime odd pairs $(b, d) \leq L$, solving (7.2) at L . Its real solution interval has length

$$1 - \frac{(L-b)(L-d) + (M-L^2)}{bd} < 1;$$

strictness at a square uses $b, d \leq L - 1$. There is at most one integer solution (a, e) per pair. Put its event at $\tau = \max(a, e, b, d)$ and let j_t count such events. The central masses are $\mathbf{1}_{t=1} + 2p_t - j_t$, $1 \leq t \leq L$: a collision is subtracted exactly when both witnesses have become available. For $N = 1, 2$, $L = B$ and this is the entire histogram.

For $N \geq 3$, $K < B$. Construct (7.6)'s open holes once, at $T = K$. Every survivor $R_N(r) > K$ has one hole about a reduced even fraction a/q . Put $d = |aM - qr|$, and let u be the least positive residue of $a^{-1} \bmod q$ on the left (including equality), or $-a^{-1} \bmod q$ on the right. It is odd. With its side neighbor b_0 at K , membership gives

$$db_0 < M - qK, \quad b_0 > K - q, \quad qK < M.$$

Hence $q < K$, $d < K$ (if $d > 0$, use $(M - qK)/(K - q) \leq K$), $d^2 < M$, and $x = M/(q + d) > b_0 \geq u$. Positive consecutive members of $u + q\mathbb{Z}$ straddling x are

$$b_- = u + q \left\lfloor \frac{M - u(q + d)}{q(q + d)} \right\rfloor, \quad b_+ = b_- + q.$$

Each gives an integral determinant-one witness of height $f(b) = \max(b, |M - db|/q)$, since $bd \equiv M \pmod{q}$. Below x the function is $(M - db)/q$, decreasing or constant; above x it equals b up to M/d , after which both branches increase. Also $f(b_+) = b_+$: for $d > q$ use

$$(d - q)(x + q) - M = \frac{q(d^2 - q^2 - 2M)}{d + q} < 0;$$

for $d \leq q$ this is immediate. Thus the progression minimum is

$$h = \min\{(M - db_-)/q, b_+\}. \quad (8.1)$$

These witnesses imply $K < h \leq f(b_0) \leq M/q \leq B$. For any integer $K \leq T < h$, $qT < M$ and $T < b_+$, so the largest side denominator at threshold T is at most b_- . Its numerator cost exceeds T . The residue remains inside this cusp's open hole; the global comparison in Theorem 7.1's proof excludes all other fractions. Hence $R_N(r) = h$. For $d = 0$ this gives M/q .

Insert each survivor at (8.1). If $K = L + 1$, insert the bridge mass $M - S_M(K) - C_M(L)$ at K ; if $K = L$ the central part already contains it. There is no skipped integer between L and K . Central residues, bridge and survivors exhaust M . There are $O(M)$ central pairs and initial even fractions. Gcds and inverses use $O(\log M)$ Euclidean arithmetic steps each. Cache the inverse once per cusp; at most M surviving integers in disjoint holes then need constant arithmetic work each. Dense bins and cumulative sums need $O(M)$ additions and storage slots. $\square$

This cost is exponential in N and is not a bit-complexity theorem or a timing comparison with all alternatives. The companion's batched route uses no defining minimum oracle or all-threshold rescan.

9 The marked lattice limit

We prove the discrete-to-continuous bridge explicitly. All vectors are rows. Set

$$G = \mathrm{SL}_2(\mathbb{R}), \quad \Delta = \Gamma^0(2) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : b \equiv 0 \pmod{2} \right\}, \quad X = \Delta \backslash G,$$

$$\mathcal{A} = \{(k, b) \in \mathbb{Z}^2 : b \text{ odd}\}, \quad \mathcal{A}_{\mathrm{pr}} = \{w \in \mathcal{A} : \gcd(w_1, w_2) = 1\}.$$

$$h(x) = \begin{pmatrix} 1 & 0 \\ -x & 1 \end{pmatrix}, \quad a(Q) = \mathrm{diag}(Q, Q^{-1}).$$

The stabilizer of $\mathcal{A}$ and of its primitive subset is Δ . The last coordinate of $(k, b)\gamma$ is $k\gamma_{12} + b\gamma_{22}$; preservation of both primitive parity classes forces $\gamma_{12} = 0$, $\gamma_{22} = 1$ modulo two, and these conditions suffice. Reduction modulo two shows index three. Use Haar probability μ_Δ on X . The horocycle period is one, since $h(x) \in \Delta$ exactly for integral x .

The external inputs are Heersink's continuous horosphere theorem [2, §2, pp. 3–4; Theorem 4, p. 7], valid for every finite-index subgroup, and Howe–Moore [3, Theorem 1.1, p. 2; Definition 2.6, p. 4]. The first, in dimension one with source torus $\mathbb{R}/\mathbb{Z}$ and both measures of mass one, gives

$$\int_0^1 f(\Delta h(x)a(Q))dx \longrightarrow \int_X f d\mu_\Delta \quad (9.1)$$

for bounded continuous f ; its joint form retains the source coordinate. The second applies to connected noncompact simple real Lie groups with finite center. Matrix coefficients of a strongly continuous unitary representation without invariant vectors vanish at infinity. No discrete marked theorem is imported. The following smearing argument uses the method of Marklof [4, §4, Proposition 9, equations (4.8)–(4.13)].

Lemma 9.1. *For $g_{M,r} = h(r/M)a(\sqrt{M})$, $\nu_M = M^{-1} \sum_{r=0}^{M-1} \delta_{\Delta g_{M,r}}$ converges weakly to μ_Δ as $N \rightarrow \infty$.*

Proof. Conjugation gives $g_{M,r}h(1) = g_{M,r+1}$, with the last point identified with the first. Thus ν_M is exactly invariant under $T : z \mapsto zh(1)$. Smearing gives the exact identity

$$\bar{\nu}_M := \int_0^1 \nu_M \cdot h(s)ds = \int_0^1 \delta_{\Delta h(x)a(\sqrt{M})}dx, \quad (9.2)$$

by $x = (r+s)/M$. Thus $\bar{\nu}_M \Rightarrow \mu_\Delta$. Choose a compact C of smeared mass at least $1 - \varepsilon$ for all sufficiently large levels, and put $C' = \{zh(-s) : z \in C, 0 \leq s \leq 1\}$. This is compact and $\bar{\nu}_M(C) \leq \nu_M(C')$. Adding the finitely many remaining supports proves tightness. The locally compact metrizable space X therefore admits subsequential probability limits.

Since $-I \in \Delta$, the action factors through $\mathrm{PSL}_2(\mathbb{R})$, satisfying the Howe–Moore hypotheses. The regular representation on $L_0^2(X, \mu_\Delta)$ is unitary and strongly continuous, with no invariant vectors: invariance under the transitive G action forces a function to be constant almost everywhere. The elements $h(n)$ escape compact subsets; Howe–Moore gives mixing of T for Haar probability, hence ergodicity.

Let ν be a subsequential limit. Exact invariance passes to it. For bounded continuous tests, dominated convergence in s gives

$$\mu_\Delta = \int_0^1 \nu \cdot h(s)ds.$$

Each measure in this barycenter is T -invariant since $h(s)$ commutes with $h(1)$. Ergodic probabilities are extreme among invariant probabilities, so $\nu \cdot h(s) = \mu_\Delta$ almost everywhere. A countable determining family of tests gives a common full-measure set of such s . Continuity of the test integrals in s extends equality to $s = 0$. Therefore $\nu = \mu_\Delta$; tightness and uniqueness prove convergence. $\square$

Let $B_t = [-t, t] \times (0, t]$ and let $Z(g)$ be the least $\max(|u|, v)$ among $(u, v) \in \mathcal{A}_{\mathrm{pr}}g$, $v > 0$. It is positive, finite and attained. Discreteness gives positivity and attainment. For finiteness the coset $(0, 1)g + 2\mathbb{Z}^2g$ has finite covering radius, whereas the sector $v > 0, |u| < v/2$ contains arbitrarily large disks. A coset point there has odd coefficient denominator; division by its odd content gives a primitive point in some B_t . The void set $W_t = \{\Delta g : \mathcal{A}_{\mathrm{pr}}g \cap B_t = \emptyset\}$ is a Haar continuity set. Locally bounded inverses restrict possible hits in the compact closure of B_t to finitely many integer rows, none approaching zero. Discontinuity requires an image on a face, including $v = 0$. For a fixed nonzero row w , the map $g \mapsto wg$ is a submersion onto $\mathbb{R}^2 \setminus \{0\}$. Preimages of planar-null faces are Haar-null; countable union and passage to the discrete quotient preserve nullity. Finally,

$$(k, b)g_{M,r} = ((Mk - rb)/\sqrt{M}, b/\sqrt{M}).$$

With $a = rb - Mk$ this identifies the box event exactly with $R_N \leq t\sqrt{M}$. The equality $\gcd(k, b) = \gcd(a, b)$ and odd-content division preserve the congruence and mark. Lemma 9.1 and the continuity-set theorem yield

$$Z_N := R_N/\sqrt{M} \Rightarrow Z, \quad \Pr(Z > t) = \mu_\Delta(W_t). \quad (9.3)$$

10 Identification with the published Boca–Gologan law

The geometric law is already published. We identify its model and scale; its authors did not assert a finite binary-string theorem. Set

$$U = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathcal{S} = \{(q, a) \in \mathbb{Z}^2 : q - a \text{ odd}\}, \quad H = U^{-1}\Delta U.$$

Since $(k, b)U = (k - b, k)$, $AU = \mathcal{S}$. The map $\Delta g \mapsto HU^{-1}g$ preserves Haar probability and the marked void events. For $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $D_t = \text{diag}(t, t^{-1})$,

$$B_t J D_t = (0, t^2] \times [-1, 1] =: C_{t^2}.$$

Right Haar invariance gives

$$\Pr(Z > t) = \mu_H\{Hg : \mathcal{S}g \cap C_{t^2} = \emptyset\}. \quad (10.1)$$

Primitivity can be omitted: marked vectors have odd content, and division contracts these origin-centered windows.

For slopes $u \in [0, 1]$, put

$$g_\epsilon(u) = \begin{pmatrix} 1 & -u \\ 0 & 1 \end{pmatrix} \text{diag}(\epsilon, \epsilon^{-1}).$$

A row (q, a) maps to $(\epsilon q, (a - qu)/\epsilon)$. The void event in C_λ is survival past horizontal distance λ/ϵ against vertical segments of half-length ϵ at (q, a) , $q - a$ odd, starting at the omitted origin. This is BG's modulus-two model [1, Theorem 1.1(i), p. 1045; definition, p. 1044]. Its parameter is ϵq , not $2\epsilon q$.

We identify BG's angular average with Haar. Work first on $\Gamma(2) \backslash G$, whose horocycle period is two. $\Gamma(2) \subset H$ and $[H : \Gamma(2)] = 2$; projection preserves probability Haar. Conjugation by J normalizes $\Gamma(2)$ and

$$J^{-1}h(x)a(\epsilon^{-1})J = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \text{diag}(\epsilon, \epsilon^{-1}).$$

Use the two-periodic weight $w(x) = 8\mathbf{1}_{[-1,0]}(x)/(\pi(1+x^2))$, of mass one for $dx/2$. The joint Heersink theorem with this finite-index subgroup and $x = -u$ gives

$$\lim_{\epsilon \rightarrow 0} \frac{4}{\pi} \int_0^1 \mathbf{1}_{\mathcal{S}g_\epsilon(u) \cap C_\lambda = \emptyset} \frac{du}{1+u^2} = \mu_H\{\mathcal{S}g \cap C_\lambda = \emptyset\}. \quad (10.2)$$

The bounded weight is L^1 -approximable; the void boundary is Haar-null by the submersion argument. Joint continuous-test equidistribution therefore extends to this event and weight. We do not assume equidistribution on an unweighted half-period.

BG's theorem gives, for intervals $I \subset [0, 1]$ of length comparable to ϵ^c , angular survival $c_I G_2(\lambda) + O_{\delta, \lambda, 2}(\epsilon^{\theta-\delta})$, where $c_I = \int_I du/(1+u^2)$ and $\theta = \min(c + c_1, 1/2 - 2c_1)$. Partition into $\lceil \epsilon^{-1/8} \rceil$ equal cells; choose $c = c_1 = 1/8$ and $0 < \delta < 1/8$. The uniform interval error sums to $O(\epsilon^{1/8-\delta}) \rightarrow 0$, and $\sum c_I = \pi/4$. Consequently

$$\boxed{\Pr(Z > t) = G_2(t^2)}. \quad (10.3)$$

This fixed- λ source estimate identifies the law and gives no rate for the dyadic sample.

The specialization of the published survival is

$$G_2(x) = \begin{cases} 1 - 8x/\pi^2, & 0 < x \leq 1/2, \\ 1 - 6x/\pi^2 + 2H_2(x)/\pi^2, & 1/2 \leq x \leq 1, \\ 4H_3(x)/\pi^2, & x \geq 1, \end{cases} \quad (10.4)$$

$$\begin{aligned} H_2(x) &= 3x - 2 + \zeta(2) - (\ln x)^2 + 2(1-x) \ln(1/x - 1) - 2 \operatorname{Li}_2(x), \\ H_3(x) &= \operatorname{Li}_2(1/x) - (x-1) \ln(1 - 1/x) - 1. \end{aligned}$$

Products at singular endpoints are interpreted continuously. Differentiation gives the density of $\Lambda = Z^2$:

$$g_2(x) = \begin{cases} 8/\pi^2, & 0 < x < 1/2, \\ \frac{4}{\pi^2 x} \left[1 - (1-x) \ln \frac{1-x}{x} \right], & 1/2 < x < 1, \\ \frac{4}{\pi^2 x} [1 + (x-1) \ln(1 - 1/x)], & x > 1. \end{cases} \quad (10.5)$$

The tail sign follows by differentiating H_3 ; the opposite sign in BG's density display on p. 1069 is a typographical error. We import the survival and derive the density.

11 Logarithmic moments and limiting variance

Lemma 11.1. *Uniformly in N , for $z > 0$,*

$$\Pr(Z_N \leq z) \leq \min(1, 3z^2), \quad \Pr(Z_N > z) \leq \min(1, 4z^{-2}).$$

Proof. For $X \geq 1$, the number of pairs with $|a| \leq X$, odd $1 \leq b \leq X$ is at most $(2\lfloor X \rfloor + 1)\lfloor X \rfloor/2 \leq 3X^2$. Each pair determines one residue. Below one the event is empty.

For the upper tail let $\lambda(r)$ be the shortest Euclidean vector length in the full congruence lattice Λ_r . A complete Gauss-reduced basis u, v satisfies $\|u\| \leq \|v\|$, $2|u \cdot v| \leq \|u\|^2$. First order a basis so that $\|u\| \leq \|v\|$. If the scalar-product inequality fails, replace v by $v - ku$, where k is a nearest integer to $(u \cdot v)/\|u\|^2$, and reorder. Each such subtraction strictly decreases the positive integer sum of squared norms; swaps leave it unchanged. Thus the process terminates at the two displayed inequalities. They imply that u is shortest. For $ku + tv$ the cases $t = 0$, $|t| = 1$ follow from the scalar-product bound; for $|t| \geq 2$ the perpendicular component has length at least $2M/\|u\| \geq \|u\|$. Indeed the reduced angle has sine at least $\sqrt{3}/2$, so $\|u\|\|v\| \leq 2M/\sqrt{3}$. Thus $\|u\| = \lambda(r)$. The denominators of the basis generate $\mathbb{Z}$, so one is odd and its height is at most $\|v\|$. Hence $R_N(r) \leq 2M/(\sqrt{3}\lambda(r))$.

For $1 \leq Y < M$, a nonzero vector of length at most Y cannot be horizontal; orient its denominator $1 \leq b \leq \lfloor Y \rfloor$. For $g = \gcd(b, M)$ the congruence has $g(2\lfloor Y/g \rfloor + 1)$ incidences with residues. Summing gives at most $3Y\lfloor Y \rfloor \leq 3Y^2$. This includes even denominators and nonprimitive vectors. If $R_N > X$, then $\lambda(r) < 2M/(\sqrt{3}X)$. When $4M/X^2 < 1$ this cutoff is less than M ; below one the event is empty, otherwise the incidence bound gives at most $4M^2/X^2$ residues. In the remaining range the probability bound one suffices. Set $X = z\sqrt{M}$. $\square$

Theorem 11.2. *Put $\ell = \ln 2$, $L_4 = \operatorname{Li}_4(1/2)$ and*

$$\begin{aligned} Q &= 4\zeta(4) + 3\zeta(2) - \frac{7}{4}\zeta(3) + \frac{1}{2}\zeta(2)\ell^2 - \frac{7}{4}\ell\zeta(3) - \frac{1}{12}\ell^4 - 2L_4, \\ \eta_1 &= \frac{7\zeta(3)}{2\pi^2} - 1, \quad \eta_2 = \frac{4}{\pi^2}Q. \end{aligned}$$

Then

$$\mathbb{E} \log_2 R_N = N/2 + c_R + o(1), \quad c_R = \frac{\eta_1}{2\ell}, \quad (11.1)$$

$$\text{Var}(\log_2 R_N) \longrightarrow \sigma_R^2, \quad \sigma_R^2 = \frac{\eta_2 - \eta_1^2}{4\ell^2}. \quad (11.2)$$

Exact rational enclosures are

$$\begin{aligned} -0.413852654438430019863271 &\leq c_R \leq -0.413852654438430019863270, \\ 0.892302571204274625294755 &\leq \sigma_R^2 \leq 0.892302571204274625294756. \end{aligned}$$

Proof. We evaluate $\eta_j = \int_0^\infty (\ln x)^j g_2(x) dx$. For $h = 1/2$ define

$$\begin{aligned} A_j &= \int_h^1 (\ln x)^j \ln(1-x) \frac{dx}{x}, \quad B_j = \int_h^1 (\ln x)^j \ln(1-x) dx, \\ E_j &= \int_h^1 (\ln x)^j \frac{dx}{x} = \frac{(-1)^j \ell^{j+1}}{j+1}, \quad D_j = \int_h^1 (\ln x)^j dx. \end{aligned}$$

The middle contribution, multiplied by $\pi^2/4$, is $M_j = E_j - A_j + B_j + E_{j+1} - D_{j+1}$. Directly differentiable primitives for A_0, A_1, A_2 are

$$-\text{Li}_2(x), \quad -\ln x \text{Li}_2(x) + \text{Li}_3(x), \quad -(\ln x)^2 \text{Li}_2(x) + 2 \ln x \text{Li}_3(x) - 2 \text{Li}_4(x).$$

With $L_j = \text{Li}_j(1/2)$ this gives

$$A_0 = -\zeta(2) + L_2, \quad A_1 = \zeta(3) - \ell L_2 - L_3, \quad A_2 = -2\zeta(4) + \ell^2 L_2 + 2\ell L_3 + 2L_4.$$

Integration by parts with the primitive $V(x) = (1-x)(1-\ln(1-x))$ of $\ln(1-x)$ gives

$$\begin{aligned} B_0 &= -(\ell+1)/2, \\ B_1 &= 1 + \ell^2/2 + A_0, \\ B_2 &= -\ell^2(1+\ell)/2 - 2E_1 + 2D_1 + 2A_1 - 2B_1. \end{aligned}$$

The elementary values needed are $D_1 = (\ell-1)/2$, $D_2 = 1-\ell-\ell^2/2$ and $D_3 = -3+3\ell+3\ell^2/2+\ell^3/2$.

The half-values in the simplification are

$$L_2 = \zeta(2)/2 - \ell^2/2, \quad L_3 = 7\zeta(3)/8 - \zeta(2)\ell/2 + \ell^3/6.$$

Dilogarithm reflection $\text{Li}_2(x) + \text{Li}_2(1-x) = \zeta(2) - \ln x \ln(1-x)$ follows by differentiation and $x \rightarrow 0$. The same argument gives $\text{Li}_2(-x/(1-x)) = -\text{Li}_2(x) - \ln^2(1-x)/2$. These identities verify the derivative of

$$\begin{aligned} \text{Li}_3(x) + \text{Li}_3(-x/(1-x)) + \text{Li}_3(1-x) &= \zeta(3) + \zeta(2) \ln(1-x) \\ &\quad - \frac{1}{2} \ln x \ln^2(1-x) + \frac{1}{6} \ln^3(1-x). \end{aligned}$$

Both sides tend to $\zeta(3)$ at zero. At $x = 1/2$, $\text{Li}_3(-1) = -3\zeta(3)/4$ from its absolutely convergent alternating series, which proves the L_3 value. Substitution now yields

$$\begin{aligned} M_0 &= \zeta(2)/2, \\ M_1 &= \ell - \zeta(2)/2 - \zeta(3)/8, \\ M_2 &= 2\zeta(4) + \zeta(2) + \zeta(3)/4 - 2\ell - \ell^2 + \zeta(2)\ell^2/2 \\ &\quad - 7\ell\zeta(3)/4 - \ell^4/12 - 2L_4. \end{aligned}$$

On $(0, 1/2)$ the normalized contributions are $1, -\ell - 1, \ell^2 + 2\ell + 2$ for $j = 0, 1, 2$. On $x > 1$ the density has the positive expansion

$$g_2(x) = \frac{4}{\pi^2} \sum_{n \geq 1} \frac{x^{-n-1}}{n(n+1)}.$$

Tonelli and $\int_1^\infty (\ln x)^j x^{-n-1} dx = j!/n^{j+1}$ give normalized tail contributions $\zeta(2)-1, \zeta(3)-\zeta(2)+1$, and $2(\zeta(4) - \zeta(3) + \zeta(2) - 1)$. Adding and multiplying by $4/\pi^2$ proves $\eta_0 = 1$ and the stated η_1, η_2 . Log powers are absolutely integrable at zero; the positive expansion justifies their tail integrals. No quadrature is needed.

For transfer let $Y_N = \ln Z_N$. Lemma 11.1 gives $\Pr(|Y_N| > T) \leq 7e^{-2T}$ for $T \geq 0$. Layer cake, including the boundary term, gives

$$\mathbb{E}[Y_N^2 \mathbf{1}_{|Y_N| > T}] \leq 7(T^2 + T + 1/2)e^{-2T} \rightarrow 0$$

uniformly in N . Thus (9.3) transfers both log moments: $\mathbb{E}Y_N \rightarrow \eta_1/2$, $\mathbb{E}Y_N^2 \rightarrow \eta_2/4$, since $\Lambda = Z^2$. As $\log_2 R_N = N/2 + Y_N/\ell$, subtracting the squared mean proves (11.1)–(11.2). The square parameter accounts for the factor four in the variance denominator.

For the enclosures use Euler-transformed eta, $p = 2, 3, 4$:

$$(1 - 2^{1-p})\zeta(p) = \sum_{k \geq 0} \frac{\Delta^k a_0}{2^{k+1}}, \quad a_n = (n+1)^{-p}, \quad \Delta a_n = a_n - a_{n+1}.$$

Indeed $\Delta^k a_0 = \frac{1}{(p-1)!} \int_0^1 (1-x)^k (-\ln x)^{p-1} dx$ lies in $(0, 1]$; the positive geometric sum gives the eta integral with factor $1/(1+x)$. Its tail after 96 terms is in $[0, 2^{-96}]$. Use 40 terms of $\ell = 2 \sum_{j \geq 0} 3^{-2j-1}/(2j+1)$ with remainder at most $9/[4 \cdot 81 \cdot 3^{81}]$, and 96 terms of $L_4 = \sum_{k \geq 1} 2^{-k}/k^4$ with remainder at most $2^{-96}/97^4$. Set $\pi^2 = 6\zeta(2)$ exactly. Finite differences and interval operations are rational; products use all four endpoint products, squares handle zero when necessary and divisors exclude zero. Floor/ceiling rounding to 10^{24} gives the displayed endpoints, reproduced by `companion/limit_constants.py`. This certifies evaluation of the expression conditional on the analytic theorem and its named external imports. $\square$

12 The raw mean and the Jensen gap

Proposition 12.1. *For $c_{\text{raw}} = 32 \ln(1 + \sqrt{2})/(3\pi^2)$,*

$$\mathbb{E}R_N \sim c_{\text{raw}} 2^{N/2}, \quad \log_2 \mathbb{E}R_N - \mathbb{E} \log_2 R_N \rightarrow \log_2 c_{\text{raw}} - c_R > 0.$$

Proof. Including the boundary term, Lemma 11.1 gives

$$\mathbb{E}[Z_N \mathbf{1}_{Z_N > K}] = K \Pr(Z_N > K) + \int_K^\infty \Pr(Z_N > t) dt \leq 8/K.$$

Thus $\mathbb{E}Z_N \rightarrow \mathbb{E}Z$. From (10.5), the three contributions to $(\pi^2/4)\mathbb{E}Z$ are

$$A = \sqrt{2}/3, \quad B = \int_{1/2}^1 x^{-1/2} [1 - (1-x) \ln((1-x)/x)] dx,$$

$$C = \int_1^\infty x^{-1/2} [1 + (x-1) \ln(1-1/x)] dx.$$

For B substitute $y = \sqrt{x}$. A directly differentiable primitive is

$$F(y) = \frac{2y}{3} - 2 \left(y - \frac{y^3}{3} \right) \ln \frac{1-y^2}{y^2} - \frac{8}{3} \operatorname{atanh} y.$$

The singular terms cancel at one: $F(1) = 2/3 - (8/3)\ln 2$. For $a = 1/\sqrt{2}$, $F(a) = 2a/3 - (8/3)\operatorname{atanh} a$. Positive tail expansion and Tonelli give

$$C = \sum_{n \geq 1} \frac{1}{n(n+1)(n-1/2)} = \frac{8}{3} \ln 2 - \frac{2}{3}.$$

For this equality sum the partial fractions $-2/n + 2/[3(n+1)] + 4/[3(n-1/2)]$ with a common finite cutoff, using $\sum_{n=1}^K (n-1/2)^{-1} = 2H_{2K} - H_K$. Since $A = 2a/3$, the sum is $(8/3)\operatorname{atanh}(1/\sqrt{2}) = (8/3)\ln(1+\sqrt{2})$, proving the coefficient. Combine with (11.1) for the gap. Strict Jensen applies to the nonconstant limiting law, whose positive mean and integrable logarithm have been proved. There is no positive gap bound at every finite length: $N = 1$ has gap zero. $\square$

Fractional free-path moments have a methodological antecedent in Boca–Gologan–Zaharescu [6, Theorem 1.1, p. 304; Theorem 1.2 and Corollary 1.3, p. 305]. Marklof provides the logarithmic-moment and complex-transform lineage [5]. These models differ from the dyadic ensemble here; we import neither their coefficients nor their convergence rates. Tian–Qi discuss the two means and gap [7, p. 78, Table 1].

13 Scope and reproducibility

The exact mean inequality, finite law and variance formula, and evaluated limiting variance are distinct results. The large-length mean proof is closed by analytic estimates and its stated finite premises. The limiting results use named dynamics theorems and the published BG law, with uniform integrability proved here. A finite sweep proves neither equidistribution nor an asymptotic theorem. We claim no rate, short elementary formula in N , bit complexity, raw-height variance formula, or consequence for a prescribed infinite 2-adic value or Collatz trajectories.

The standalone source, PDF and companion accompany this article. The project repository provides the public reproduction entry point. This is a substantial revision of the preceding mean-only article. The companion includes claim and reproduction maps and these files:

- `companion/verify_mean.py` and `companion/finite_certificates.json`;
- `companion/distribution.py` and `companion/exact_logs.py`;
- `companion/verify_distribution.py`;
- `companion/distribution_certificates.json`;
- `companion/limit_constants.py`.

From the repository root run

```
python -B scripts/reproduce.py
python -B -O scripts/reproduce.py
python -B scripts/reproduce_distribution.py
python -B -O scripts/reproduce_distribution.py
python -B scripts/reproduce_constants.py
python -B -O scripts/reproduce_constants.py
```

Normal and optimized runs must give identical deterministic mathematical JSON. The mean checker verifies the finite premises in Section 6; the distribution checker compares a fixed finite panel and exact logarithm endpoints; the constants checker evaluates the symbolic expression by rational intervals. These checks establish implementation and certificate consistency, not universal priority or formal proof. All scientific arguments are present in this source: no unpublished research note or bibliography database is needed.

AI assistance and source limits. OpenAI GPT models assisted exploration, proof criticism, source comparison, editing and reproducibility checks. AI reviews shared exposure to the research material; they are not independent human peer review or formal verification. The author retains responsibility for arguments and attributions. The final Chen–Winterhof body remains uninspected as stated in the introduction; absence of an identified antecedent is not certification of priority.

A Directed logarithms using integer arithmetic

We specify the rounding used in (6.2). For a rational $v = u/w$ with $1 \leq v \leq 2$ let $z = (u - w)/(u + w) \in [0, 1/3]$. Then

$$\ln v = 2 \sum_{j=0}^{11} \frac{z^{2j+1}}{2j+1} + 2 \sum_{j=12}^{\infty} \frac{z^{2j+1}}{2j+1}.$$

To compute an upper, replace Sz by its ceiling, compute its square divided by S with upward rounding, start the scaled power with that ceiling, and at each step round both division by $2j+1$ and the multiplication by the scaled square divided by S upwards. Twice the twelve-term sum is an upper for the partial sum. The exact tail, using $z \leq 1/2$, is bounded by

$$\frac{2z^{25}}{25(1 - z^2)} \leq \frac{8}{3 \cdot 25 \cdot 2^{25}}.$$

Add the integer ceiling of $8S/(3 \cdot 25 \cdot 2^{25})$. For a lower, use downward rounding at every step and omit the nonnegative remainder. For $v = 1$, return exactly zero. These rules give a lower ℓ_2^- and upper ℓ_2^+ for $S \ln 2$.

For an arbitrary rational $x = u/w \geq 1$, find the unique integer $k \geq 0$ with $2^k w \leq u < 2^{k+1} w$, by integer bit lengths and a comparison. Apply the unit-interval upper to $u/(2^k w)$ and add $k\ell_2^+$. Every operation is an integer comparison, product, floor or ceiling. Since all rounding maps are monotone on nonnegative inputs, the computed scaled powers bound the true scaled powers by induction. The uniform tail bounds the true analytic remainder; it does not assume that an upward-rounded approximation of z still equals z . This proves the directed inequalities used by every summand of the finite bridge.

References

- [1] F. P. Boca and R. N. Gologan, On the distribution of the free path length of the linear flow in a honeycomb, *Annales de l'Institut Fourier* **59**(3) (2009), 1043–1075. doi:10.5802/aif.2457. Published full text. Theorem 1.1(i), p. 1045, and model/parameter definitions, p. 1044.
- [2] B. Heersink, Equidistribution of Farey sequences on horospheres in covers, arXiv:1712.03258v2. arXiv version 2. Definitions in §2, pp. 3–4, and Theorem 4, p. 7 of the identified text.
- [3] C. Ciobotaru, A unified proof of the Howe–Moore property, arXiv:1403.0223v2. arXiv version 2. Theorem 1.1, p. 2, and Definition 2.6, p. 4. The classical antecedent is R. Howe and C. C. Moore, Asymptotic properties of unitary representations, *Journal of Functional Analysis* **32** (1979), 72–96, doi:10.1016/0022-1236(79)90078-8.
- [4] J. Marklof, Smallest denominators, arXiv:2310.11251v2. arXiv version 2. §4, Proposition 9, equations (4.8)–(4.13).
- [5] J. Marklof, The log moments of smallest denominators, *Integers* **24** (2024), A55. Published full text.

- [6] F. P. Boca, R. N. Gologan and A. Zaharescu, The average length of a trajectory in a certain billiard in a flat two-torus, *New York Journal of Mathematics* **9** (2003), 303–330. Published full text. Theorem 1.1, p. 304, Theorem 1.2 and Corollary 1.3, p. 305.
- [7] T. Tian and W.-F. Qi, Expected values for the rational complexity of finite binary sequences, *Designs, Codes and Cryptography* **55** (2010), 65–79. doi:10.1007/s10623-009-9331-x.
- [8] Z. Chen and A. Winterhof, Probabilistic results on the 2-adic complexity, arXiv:2501.16785v1, 28 January 2025. arXiv version 1. Detailed theorem locators in this manuscript refer to this version.
- [9] Z. Chen and A. Winterhof, Probabilistic results on the 2-adic complexity, *Designs, Codes and Cryptography* **93** (2025), 2191–2203. doi:10.1007/s10623-025-01592-1. Version of record identified; detailed theorem locators above refer to arXiv v1.
- [10] H. Lin, L. Xiao and Z. Chen, A brief view on the linear and 2-adic complexities of pseudorandom binary sequences, *AIMS Mathematics* **11**(7) (2026), 20267–20290. doi:10.3934/math.2026823.
- [11] Y. Ren and A. Winterhof, Symmetric measures of pseudorandomness for binary sequences, arXiv:2603.23166v1, 24 March 2026. arXiv version 1.
- [12] M. Goresky and A. Klapper, *Algebraic Shift Register Sequences*, Cambridge University Press, 2012. doi:10.1017/CBO9781139057448. Growth lemma cited as Lemma 18.5.1 in [8]. Public author draft, 14 October 2009, Section 21.5.a, Lemma 21.5.1: author draft. Draft and published numbering are not identified here.
- [13] J. W. S. Cassels, *An Introduction to the Geometry of Numbers*, Springer, 1997 reprint, Chapter III, Section 1 (Minkowski’s convex body theorem). doi:10.1007/978-3-642-62035-5_4.